Generative AI for RAG-Based AI System for Automated Telecom Customer Support

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The Telecom Customer-Support Background and Challenge

Why context-aware answers matter

Background->

Telecom customer support handles a high volume of diverse and complex queries related to billing, plans, network issues, roaming, device compatibility, and service outages. Traditional support systems and standalone Large Language Models (LLMs) often struggle to provide accurate, context-aware, and verifiable responses due to the dynamic nature of telecom policies, large knowledge repositories, and the need for technical, real-time, and evidence-based information.

This research addresses the challenge of generating accurate, context-aware, and reliable telecom customer support responses using Retrieval-Augmented Generation (RAG) with Generative AI.

Challenge->

Traditional Support

- Rule-based and keyword systems struggle with complex or ambiguous queries.
- Large knowledge repositories are difficult to search consistently.
- Customers may need repeated interactions.

Standalone LLMs

- May generate inaccurate or outdated information.
- Can answer without evidence from current telecom policies or technical documents.
- Hallucinations create service and trust risks.

Business Impact

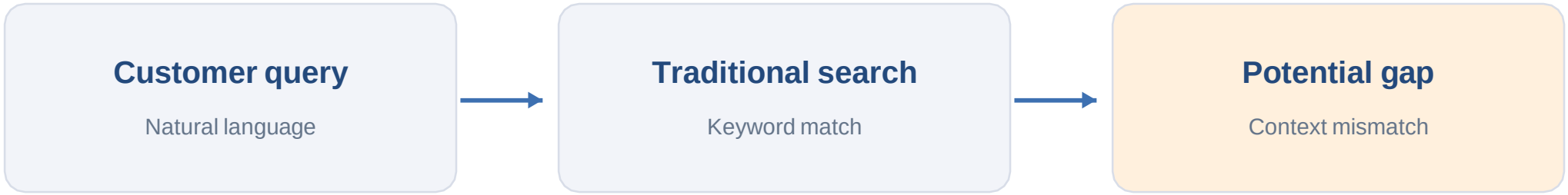
- Longer waiting and resolution times.
- Inconsistent guidance on billing, roaming, plans and technical issues.
- Need for grounded, verifiable responses.

Research challenge: generate accurate, context-aware telecom support responses.

Problem Statement

The research motivation

Telecom support requires answers that reflect the correct service, product, policy and technical context. Keyword retrieval may miss meaning, while a standalone generative model may produce unsupported content.



Research response: retrieve trusted context before generating an answer.

Literature Review: Evolution of Support AI

The thesis reviews progression from rules to knowledge-grounded generation

Rule-based & keyword

• Transparent for simple queries; limited semantic and conversational context.

Traditional ML / NLP

• Useful for classification and intent detection; relies on labelled features.

Deep learning

• Learns richer text representations but may depend on static training data.

Transformers & LLMs

• Strong language understanding and generation; can still hallucinate.

RAG systems

• Retrieve external evidence and condition generation on relevant context.

Key direction: combine retrieval with generation for evidence-grounded support.

Aim and Research Objectives

From data preparation to comparative evaluation

AIM

Design, develop and evaluate a RAG-based Generative AI framework for automated telecom customer support.

01

Data selection

Collect and filter customer-support records for telecom-specific experiments.

02

Preprocessing

Clean text, construct customer–agent QA pairs and split by conversation.

03

Retrieval framework

Compare TF-IDF, dense vector retrieval, RAG and optimised hybrid RAG.

04

Evaluation

Measure retrieval, response-quality, grounding, latency and composite performance.

05

Responsible AI

Identify privacy, reliability, governance and deployment considerations.

Research Gap and Motivation

The thesis focuses on domain-specific retrieval and evaluation

Domain grounding

- Generic responses may miss telecom-specific product and issue context.
- Retrieval must prioritise relevant service information.

Retrieval optimisation

- Sparse and dense retrieval have different strengths.
- Metadata filtering and reranking may improve relevance.

Evaluation quality

- Retrieval and answer metrics need to be considered together.
- Proxy measures require careful interpretation.

Study contribution: a reproducible comparison of four telecom support retrieval/generation configurations.

Methodology: End-to-End Experimental Workflow

Offline quantitative experiment; conversation-level split and fixed seed



Scope boundary: offline query resolution only, no live account changes or network operations.

Experimental Configurations

Four configurations evaluated using the supplied model_comparison.csv

1 **Optimised Hybrid RAG**

Sparse + dense retrieval, metadata filtering, reranking and generation

2 **Traditional Retrieval**

TF-IDF lexical retrieval baseline

3 **Vector Retrieval**

BGE-M3 embeddings indexed with FAISS

4 **Standard RAG**

Retrieved context passed to a generation layer

Parameters: standard Top-K = 5; candidate-K = 10; optimised Top-K = 3; random seed = 42.

Dataset and Preparation

Source counts as reported in the thesis

976,271	69,731	81,877	5,840
source records	source conversations	strict telecom records	telecom conversations

Customer–agent QA pairs

39,649 total pairs after telecom filtering



Preparation: whitespace normalisation, exclusion of extremely short messages, sequential customer-to-agent pairing, and conversation-level 80:20 split using random state 42.

Proposed Telecom RAG Architecture

Hybrid retrieval, metadata filtering and reranking before answer generation

Customer query
Normalise text

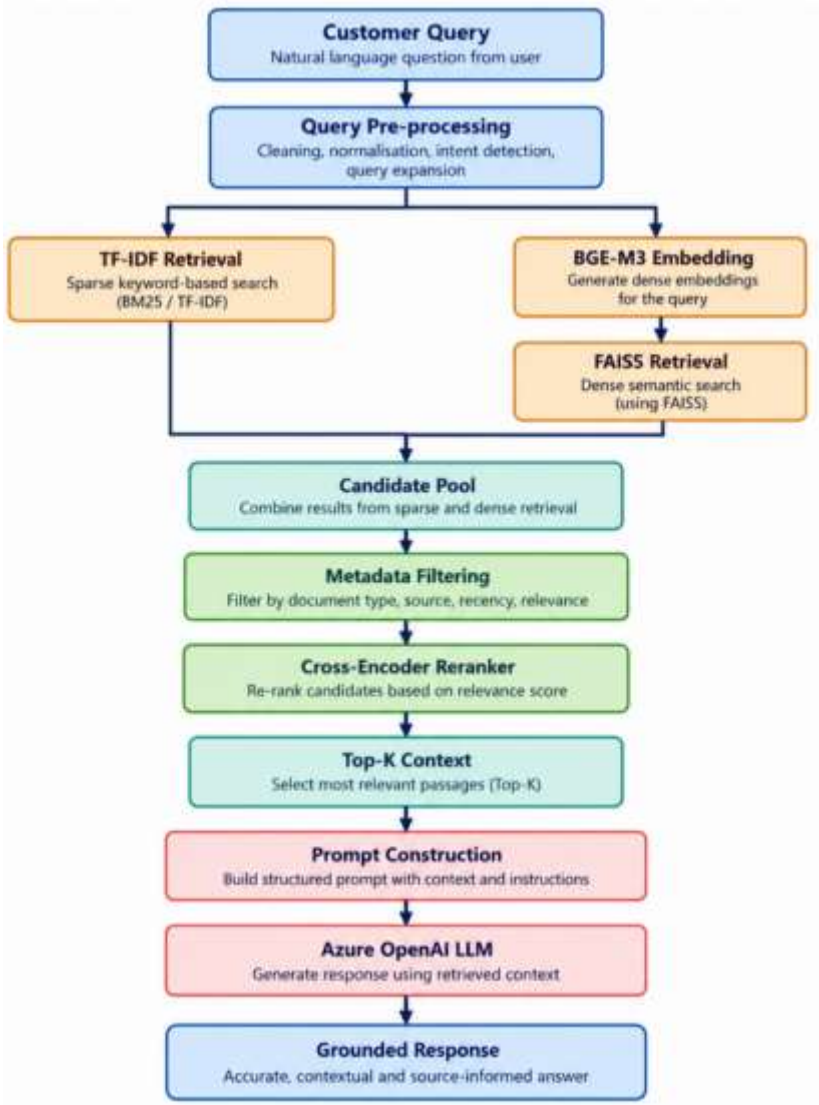
Candidate retrieval
TF-IDF + BGE-M3

Vector index
FAISS

Filter & rerank
Metadata + cross-encoder

LLM generation
Azure OpenAI configured

Evaluation
Accuracy proxies, relevance, grounding



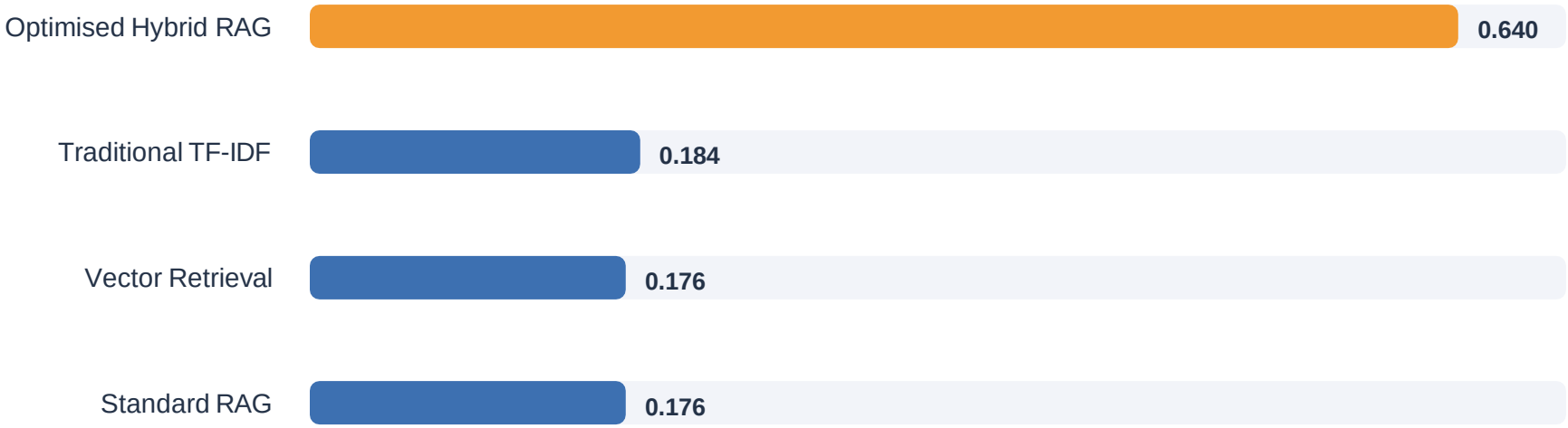
Evaluation and governance layer

Check context relevance, response quality, latency, evidence grounding and privacy boundaries.

Key Results: Retrieval Effectiveness

Reported proxy metrics from the supplied aggregate CSV

Precision@K



Interpretation

Optimised Hybrid RAG records the highest reported Precision@K (0.640) and MRR_proxy (0.740). Recall@K_proxy values are extremely small and should be checked against the implementation's denominator and relevance-label logic.

Limitations and Future Work

Important boundaries on interpretation and next steps

Current limitations

- Synthetic/noisy source data rather than authentic telecom interactions.
- Metadata-based relevance labels, not expert annotations.
- Lexical F1, embedding similarity and lexical grounding are proxies.
- Non-monotonic timestamps do not support validated resolution-time claims.
- Missing raw queries and sample-level outputs; results remain provisional.

Recommended next steps

- Create human-annotated telecom relevance and answer-quality benchmarks.
- Expand evidence sources with validated FAQs, operator policies and 3GPP standards.
- Evaluate factuality and entailment; audit hallucinations with experts.
- Tune hybrid retrieval, candidate size, reranking and latency/cost trade-offs.
- Pilot with anonymised real data and human-in-the-loop escalation.

Future directions: multilingual support, real-time knowledge updates and Agentic AI.

Key Results: Response Quality, Grounding and Latency

Read reported metrics with their operational definitions in mind

Response quality

- Optimised Hybrid RAG Answer_Similarity_F1: 0.0330
- Vector Retrieval and RAG: 0.0277
- Response relevance is also a proxy.

Grounding

- RAG and Optimised RAG score 1.00 on lexical overlap.
- Lexical overlap does not prove factual entailment or eliminate hallucinations.

Latency

- TF-IDF: 52.597 ms
- Vector Retrieval / RAG: about 8.695 seconds
- Optimised Hybrid RAG: about 8.692 seconds

No single metric is sufficient

Retrieval quality, response quality, grounding and latency must be assessed together; the aggregate results are provisional pending a validated rerun.

Conclusion and Contribution

Summary of the thesis findings

Framework

Developed an offline telecom RAG workflow combining retrieval, evidence context and LLM response generation.

Comparison

Evaluated TF-IDF, vector retrieval, standard RAG and optimised hybrid RAG.

Best reported score

Optimised Hybrid RAG achieved the highest composite score (0.700056) in the supplied CSV.

Caution

Proxy-based evaluation and missing run artefacts mean results need rerunning and independent validation.

Practical direction

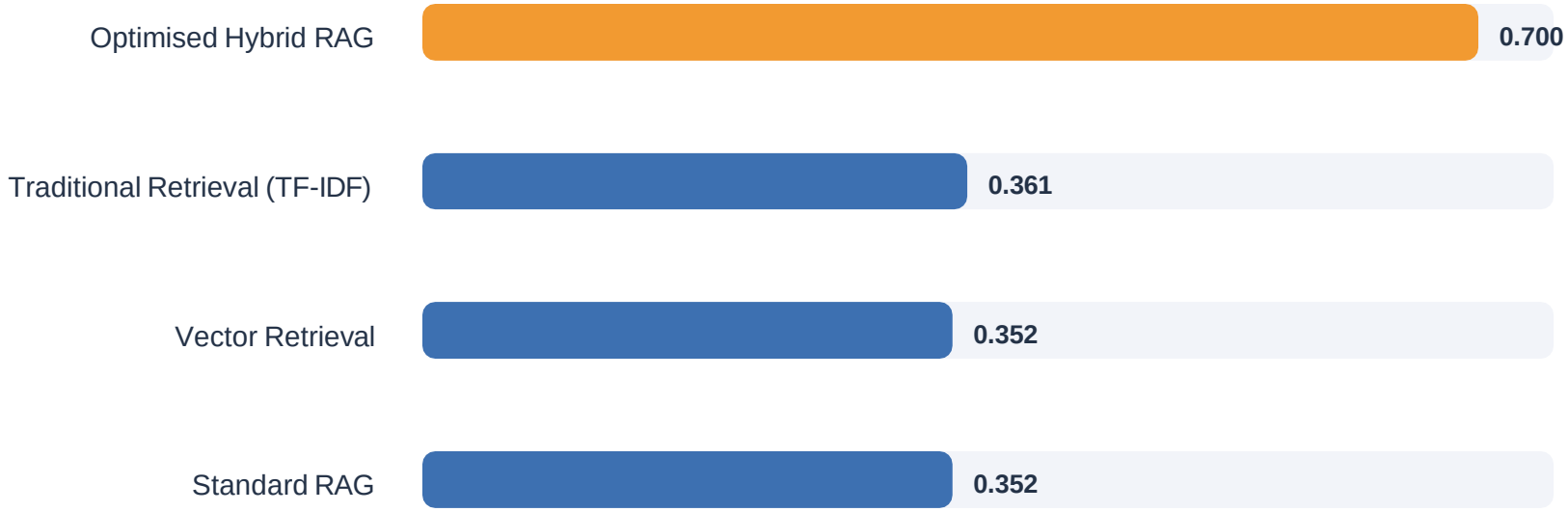
Use source-aware answers, confidence thresholds, human review and controlled knowledge updates.

A promising experimental foundation—not yet evidence of production-level performance.

Analysis->Composite Score and Overall Ranking

Weighted composite score reported by the experiment

Optimised Hybrid RAG achieved the highest composite score (0.700), outperforming TF-IDF (0.361), Vector Retrieval (0.352), and Standard RAG (0.352). These results suggest that combining retrieval techniques improves performance for telecom customer support. However, further experimental validation and statistical significance testing are required to confirm the robustness of these findings.



Main finding

Optimised Hybrid RAG ranks first on the supplied composite score. Vector Retrieval and Standard RAG have identical aggregate values in the CSV, and missing run-level artefacts limit independent verification.

Thank You